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1- HEAD AND BODY SPIKE NOISE

1-1 Overview

This test detects small intermittent spike noise sources that may be present in either the RF-shielded room and/or the MGD/RRF.

Note

The Head and Body Spike Noise procedure is to be performed on all systems.

Short time duration noise spikes generate broadband frequency interference patterns (i.e., corduroy or tweed-like patterns, also horizontal, vertical, or diagonal stripes) artifacts in MR images. Past experience has shown spike noise to be due to loose connectors, intermittent ground loops, faulty electronic components in coil assemblies, and corruption of digital data in digital storage or processing areas.

Spike noise greater than $0.5 N_{env}$ above the random noise peaks measured at the UTNS J30 receiver 1 sample port can cause image artifacts. A system entirely absent of spike noise is achievable and is the norm. The test consists of the following:

- The head coil is placed in the bore without a phantom.
- A scan is set up, prescan is started and paused to set the MGD/RRF in the receive mode. The receive noise is then observed at the UTNS J30 receiver 1 sample port with a storage scope. This configuration checks for external noise. Then the patient comfort fan is turned on, and the receive noise is again observed, to check for noise originating from the patient fan. Then the fan is turned off.
- Prescan is restarted and the receive noise is again observed at the UTNS J30 receiver 1 sample port with a storage scope. This configuration checks for noise caused by vibration.
- This sequence is repeated seven more times with scan protocols for different axes and frequency directions.
- After the last sequence (Body Axial), the Body SNR or TLT Sphere and Loader are installed, a scan is taken, and the images are reviewed for the presence of hot pixels or artifacts.

1-2 Required Tools

Item	Description	Part Number	Qty
1	100 mHz storage oscilloscope, Tektronix® 468 or equivalent	46-183029P61 or 46-183029P64	1 1
2	TPS RF Service Interface Kit	46-301927G1	1
3	SMB-femal to BNC-male Test Cable (usually in item 2, above)	46-301549P5	1
4	Body TLT Sphere Phantom	46-265635G6	1
5	Long Body Loader or SPT Body Loader	46-287902G1 2135652-2	1 1

WARNING!

POISON HAZARD! THE PHANTOM CONTAINS NICKEL, A SUSPECT CARCINOGEN. DO NOT INGEST. DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.

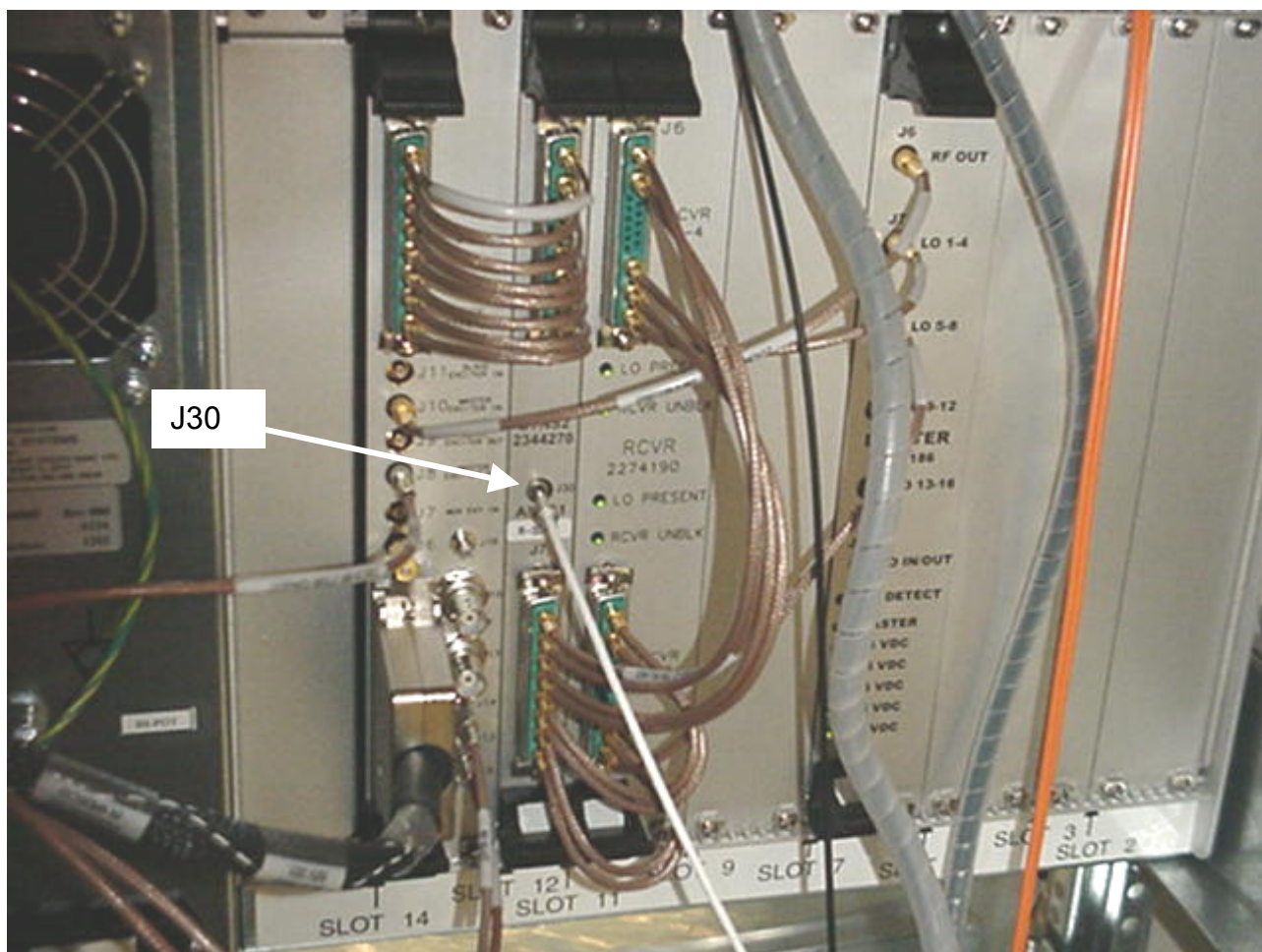
1-3 Hardware Preparation for Spike Noise Data Collection

1. Remove the front panel from the Systems Cabinet.
2. Disable the RF Output from the RRF Master Exciter by setting the RF Out toggle switch (located on the RRF front panel I/O board) to the **Disable** position.
3. Bypass the head and body TNS out of the circuit using a BNC bullet.
4. Connect the oscilloscope to J30 on the RRF UTNS.

Note

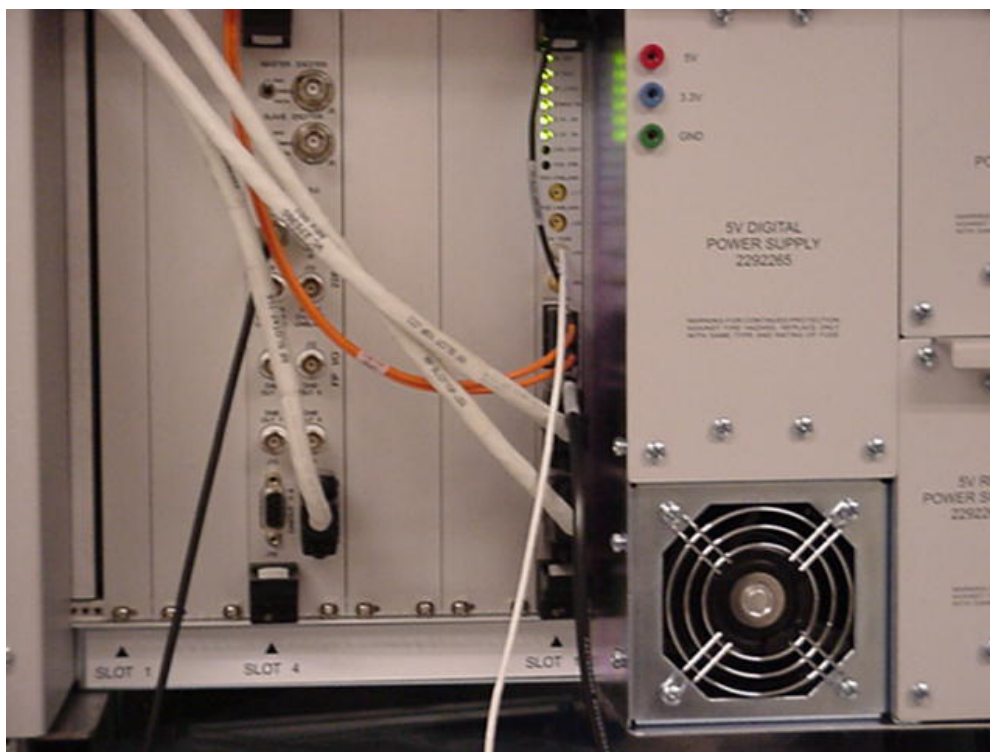
Use the Systems Cabinet service outlet for oscilloscope power to prevent any unwanted ground loops that could cause higher floor noise.

- a. Channel 1 to J30 on the RRF UTNS using test cable 46-301549P5. (See Illustration 1-1.)



LOCATION OF UTNS J30 RECEIVER
ILLUSTRATION 1-1

- b. Channel 2 to Receiver, DATA IN WINDOW. See Illustration 1-2.
- c. External Trigger to J7 Scope Trigger on the RRF Front Panel I/O board.



SPIKE NOISE POWER SUPPLY
ILLUSTRATION 1-2

4. Set up the oscilloscope as follows:
 - a. Channel 1 - 5 mV/div, 1 m Ω input
 - b. Channel 2 - 5 V/div, 1 m Ω input
 - c. Time Base - 5 msec/div
 - d. Trigger - auto, external

1-4 Spike Noise Scan Preparation

1. At the operator workspace, select the Scan icon on the desktop control panel.
2. If necessary, exit out of any previous exams by selecting **[End Exam]**.
3. Click on **[New Pt]** and enter the following:
ID: **geservice**
Name: **spike noise**
Weight (lb.): **111**
4. Remove all phantoms from the patient table. This is an empty bore test.
5. Place the head coil without a phantom on the cradle. Landmark on the center of the head coil. At the keypad on the front magnet enclosure, press LANDMARK, and MOVE TO SCAN.

*The following three steps are **proprietary** and only available for GE use, and to sites with a valid Advanced Service Package Limited License. The non-proprietary procedure is listed after these steps.*

6. Set Patient Protocols to **Service**.
7. In the Protocol field, type **o.3.1** (o=Other, 3=protocol number, 1=series number) and press **<Enter>**.
or
Click on **Other** and select protocol **3** and series **1** from the menu.
8. Click on **[Accept]** to load the protocol.

Non-proprietary procedure:

At the operator workspace, prepare the system for a "Spike Noise - Head, Axial" scan using the scan protocol (**o.3.1**) shown in the "Service Protocols" procedure located on the service methods CD-ROM.

9. Click on **[Save Series]** and then **[Prepare to scan]**.

Note

Use Series **2-8 (o.3.2 - o.3.8)** per later procedure steps.

TABLE 1-1
SERIES 1 - 8 SCAN PARAMETERS

Series 1:[Head] [Axial] I60 S60 [A/P]	Series 5:[Body] [Coronal] P60 A60 [R/L]
Series 2:[Head] [Axial] I60 S60 [R/L]	Series 6:[Body] [Sagittal] R60 L60 [S/I]
Series 3:[Head] [Sagittal] R60 L60 [A/P]	Series 7:[Body] [Axial] I60 S60 [R/L]
Series 4:[Head] [Coronal] P60 A60 [R/L]	Series 8:[Body] [Axial] I60 S60 [A/P]

1-5 Sample Noise Data Collection and Analysis

1. Click on **[Manual Prescan]** and set R1=**11**, R2=**15**, TG=**0**.
2. Press the PAUSE SCAN button on the console keyboard. Press the FAN ON button on the magnet enclosure to turn off the patient comfort fan (the fan turns on automatically with scan software).

Note

Ignore the first 30 seconds of data.

The MGD/RRF are now in the receive mode and any noise can be observed at the UTNS J30 receiver 1 port. Scope must be in auto trigger mode.

3. Put the scope in continuous acquisition mode. The method for doing this will vary with scope type. Clear the oscilloscope display and ensure the trigger is in the auto position.

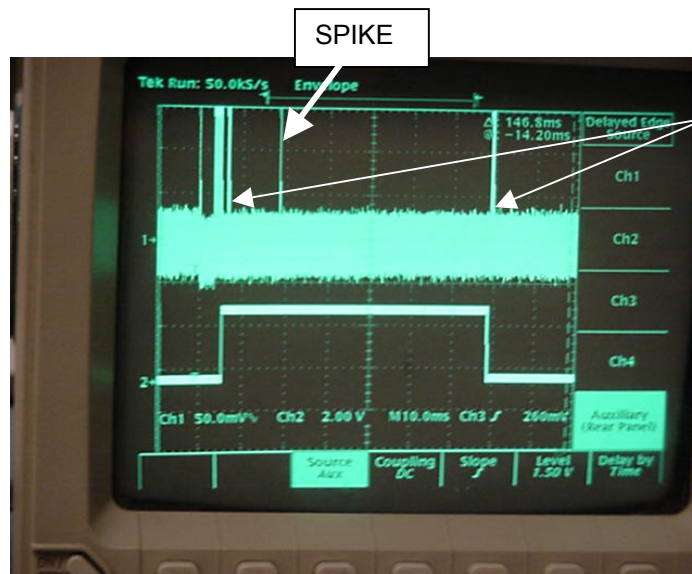
- Observe the scope display for about two minutes. The noise envelope (channel 1) is typically 45-50 mV p-p. Verify that there are no spikes $>0.5 N_{env}$ (where N_{env} is the noise envelope peak to peak value) above the noise envelope. See Illustration 1-3.

NOISE SPIKE IS BELOW
SPEC BUT WILL
PROBABLY GET WORSE

CH1 (A/D SAMPLE)
SENSITIVITY =
50 mV/DIV (1.5T)
10 mV/DIV (1.0T)

TIME BASE=5 ms/DIV
TRIGGER= AUTO,
EXTERNAL

CH2 (DATA IN WINDOW)
SENSITIVITY=5 V/DIV



TRANSITION SPIKE IS
OKAY

SPIKE SPEC:
 $1.5T = 0.5N_{env}$

SPIKE NOISE SCOPE WAVEFORM (PAUSED PRESCAN)
ILLUSTRATION 1-3

Note

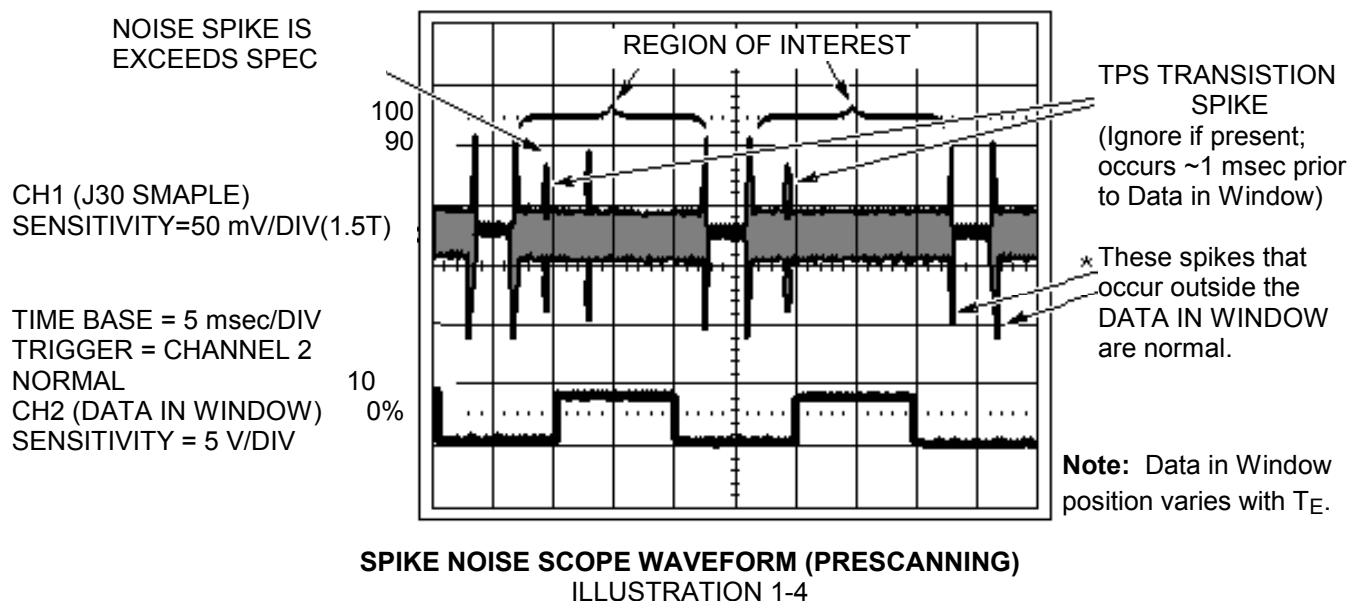
Ideally, there should be no spikes beyond the envelope. Any spikes present indicate that a connection is loose, or a component is bad (e.g., intermittent connection of power cable to a shield cooler cold head, or copper fingers on exam room door are damaged). Noise spikes that are greater than the spike spec create images with corduroy-like artifacts. Noise spikes that are presently less than the spike spec will probably get worse over time.

Note

If there are no spikes present, and you are in doubt whether the setup is working, you can simulate an RF leak by opening the exam room door, or by turning the room lights on or off if the site is RF quiet. Clear the oscilloscope display and observe the new waveform on the display. You should now see noise spikes beyond the waveform envelope. Don't forget to close the exam room door when you are finished.

- Press the FAN ON button on the magnet enclosure to turn on the patient comfort fan. Clear the oscilloscope display and ensure that the trigger is in the Auto position. Observe the oscilloscope display for about two minutes. The noise envelope (channel 1) should remain as described in step 4 above.

6. Clear the oscilloscope display and change the trigger from Auto to Normal.
7. Press the FAN ON button on the magnet enclosure to turn off the patient comfort fan.
8. Press the START SCAN button on the console keyboard to resume prescan. Click on **[Scan T/R]**.
9. Observe the scope display after two minutes. The noise envelope (channel 1) should be approximately 50mV. Verify that there are no spikes $>0.5 N_{env}$ (where N_{env} is the noise envelope peak-to-peak value) above the noise envelope in the region of interest (except for the beginning and end transition spike). See Illustration 1-4.



Important! The Region of Interest extends beyond when the observed Data In Window signal is high because the Data In Window position varies with T_E . The Transition Spike (not always present) occurs when the RF subsystem transitions from transmit to receive frequency; it occurs ~1 msec prior to the Data In Window. (The frequency transition point can be confirmed by viewing the DDS OUT signal on the Exciter Board.)

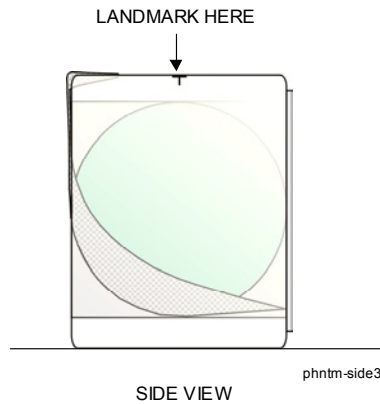
Note

If the spike noise appears only during prescan, the problem is more likely to be a loose connector, freely hanging cables near the magnet, poorly soldered components, or coil mounting hardware. Anything that is subject to vibration from gradient activity should be investigated.

10. Click on **[New Series]**. Set up scan prescription for series #2 scan per notes and Table 1-1 in Section 1-4, then repeat steps 7 through 9 of Section 1-4.
11. Repeat the previous step until all head prescans (series #1 through #4) have been completed.

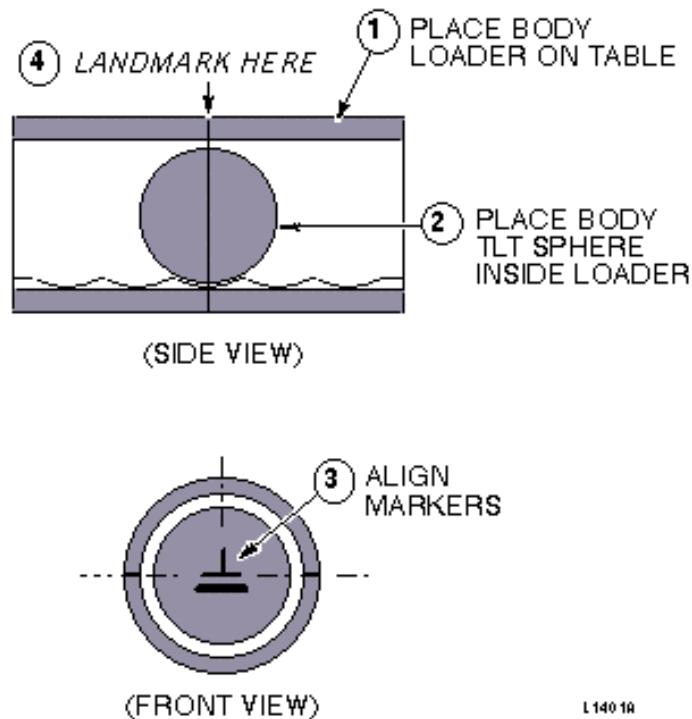
1-6 Image Data Collection/Analysis

1. Disconnect the oscilloscope from J30 on the RRF UTNS and restore all cables to their normal configuration.
2. Reset the RF Out switch on the RRF Front Panel I/O board to the **Enable** (left) position.
3. Install the front cover panel on the Systems Cabinet.
- 4a. **Setup using SPT Loader:** Position the SPT body loader and TLT sphere on the table and landmark per Illustration 1-5A.



SPT BODY LOADER AND TLT PHANTOM SETUP
ILLUSTRATION 1-5A

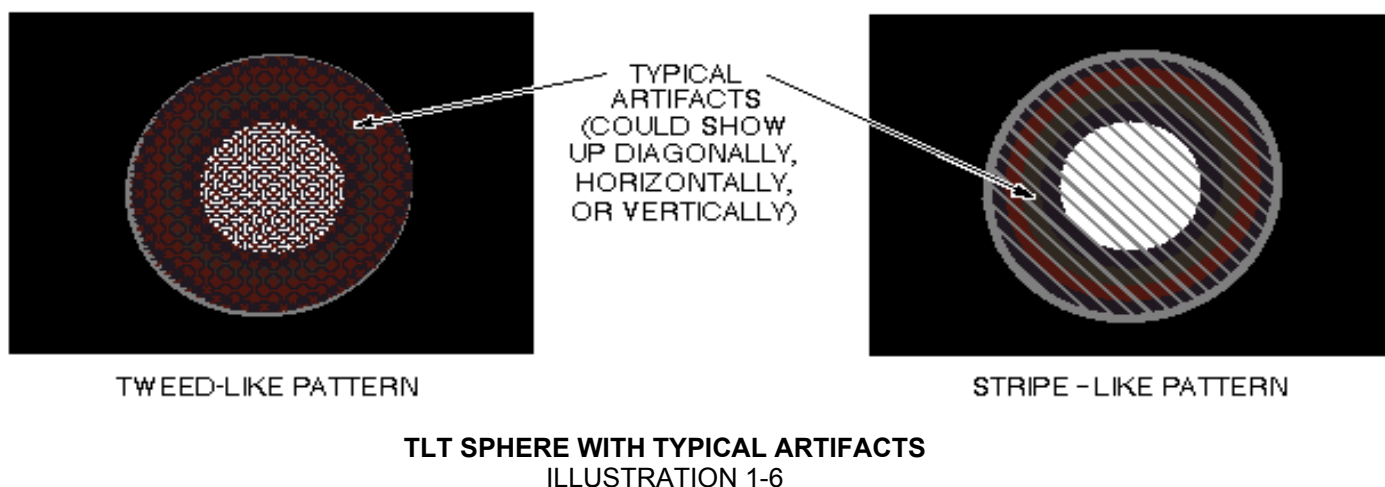
- 4b. **Setup using Long Body Loader.** Position the body TLT sphere in the center of the long body loader and landmark per Illustration 1-5B.



LONG BODY LOADER AND TLT PHANTOM SETUP
ILLUSTRATION 1-5B

5. Click on **[New Series]** and select the last scan prescription from the procedure in Section 1-4.
6. At the front enclosure, press LANDMARK, then MOVE TO SCAN.
7. Click on **[Save Series]**, then **[Prepare to Scan]**.
8. Click on **[Scan]** (system auto prescans first).

9. Display and examine each of the twenty images for tweed-like or stripe-like artifacts. See Illustration 1-6. If artifacts are present, use the procedures described in the *EPI White Pixel Troubleshooting Guide* to troubleshoot the system.



REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Feb. 4, 2003	C. MacDonald	New Spike Noise Checks doc for MGD based on edits of original by Rudy Kopp. Replaced original Illustrations 1-1, 1-2, 1-3 and removed Section 2.
1			
2			
3			
4			
5			